CSG2001: AUGMENTED & VIRTUAL REALITY

Augmented Reality: Principles, Working & Applications

Comprehensive study of spatial tracking, core rendering mechanics, and industrial use-cases

PRESENTED BY GROUP MEMBERS

Member 1 Member 2 Member 3 Member 4 Member 5

FOUNDATIONAL CONCEPT

What is Augmented Reality (AR)?

Digital Enhancement

Adds digital elements to a live view using smartphone cameras or smart glasses.

Spatial Integration

Blends computer-generated assets seamlessly into the physical user environment.

Ubiquitous Access

Primarily accessed via everyday mobile devices or advanced standalone HMDs.

First coined in 1992 by Boeing engineers Thomas Caudell and David Mizell for airplane wire-assembly aids.

SYSTEM ARCHITECTURE

How AR Works: Working Principles

STEP 01

Sensor Input

Cameras, microphones, and accelerometers capture environmental data.

STEP 02

Context Analyzer

Scans, segments, and tracks environmental elements and 3D spatial positions.

STEP 03

MAR Execution Engine

Simulates actions, verifies designer rules, and computes digital render logic.

STEP 04

Display Output

Projects integrated 3D assets to end users via screens or optical combiners.

Computer vision bridges physical sensor inputs with digital execution engines in real-time.

REALISM MECHANICS

Placing, Scaling & Occlusion

Placing & Anchoring

Stationary AR objects stick to specific surfaces (floors, walls) and remain fixed even when users move.

Scale & Perspective

Assets adjust size and orientation dynamically based on distance to mimic real-world depth and physics.

Occlusion

Virtual objects correctly hide behind physical obstacles, maintaining immersion without breaking realism.

Proper lighting estimation and solid rendering prevent digital items from appearing to float in mid-air.

ENVIRONMENTAL UNDERSTANDING

Tracking in AR: Inside-Out vs. Outside-In

Inside-Out Tracking

Cameras and sensors are built directly into the device body.

- ✓ Examples: Smartphones, Microsoft HoloLens

Outside-In Tracking

Sensors or cameras are mounted externally on walls or stands around the room.

- ✓ Example: HTC Vive headset setups

Simultaneous Localization and Mapping (SLAM) and ARCore feature-points calculate exact pose and orientation.

CORE CLASSIFICATION

Marker-Based vs. Marker-Less AR

Marker-Based AR

Requires a specific visual trigger such as a QR code or reference image to render 3D models.

High recognition accuracy under controlled visual setups.

Marker-Less AR

Detects surface planes (floors) or uses GPS/location data to place virtual objects.

Example: Pokémon GO (location-based & plane detection)

Other variants include Projection-based AR (e.g., Google Glass) and Superimposition object tracking.

INDUSTRY IMPACT

Key Applications of AR

Education

Spatial learning lets students inspect 3D models like DNA strands directly.

Healthcare

Doctors use 3D visualization for surgical planning and diagnostics.

Retail & Shopping

Virtual try-ons and furniture placement in real homes (e.g., Lenskart, Pepperfry).

Business & Warehouses

Navigation instructions for workers and 3D architectural displays.

Also widely adopted in social media (Snapchat AR stickers) and gaming (Pokémon GO).

STRATEGIC VALUE

Why Use AR? Value & Impact

Enhanced Visualization

Translates complex 2D data and textbook concepts into interactive 3D spatial experiences.

Deep Social Engagement

Nonprofits simulate climate change impacts (droughts/floods) directly in user backyards for powerful messaging.

Operational Efficiency

Reduces error rates in surgery, assembly, and warehouse navigation with real-time contextual guidance.

AR applications drive higher user retention and practical understanding compared to traditional media.

SYSTEM LIMITATIONS

Technical Constraints: Size, Power & Heat

Power & Battery Drain

Rendering complex AR experiences and streaming vision data heavily drains device batteries.

Thermal Generation

Intensive computation generates heat, which can throttle processors within compact headset enclosures.

The 3D Design Barrier

Limited pool of creators trained in 3D design theory and computer vision modeling limitations.

⚠ Lack of established UI metaphors makes user interaction design an ongoing frontier for creators.

NEXT GENERATION

Future Scope: MR, XR & Evolution

Mixed Reality (MR)

Real-world and digital objects interact directly in real-time (e.g., Microsoft HoloLens).

Extended Reality (XR)

Umbrella term encompassing AR, VR, and MR technologies that enhance human senses.

Ubiquitous Adoption

Hardware miniaturization and SDKs like ARCore are driving widespread enterprise and consumer growth.

The virtual continuum bridges completely real environments with fully immersive virtual worlds.



Thank You

Questions & Discussion on Augmented Reality Technologies